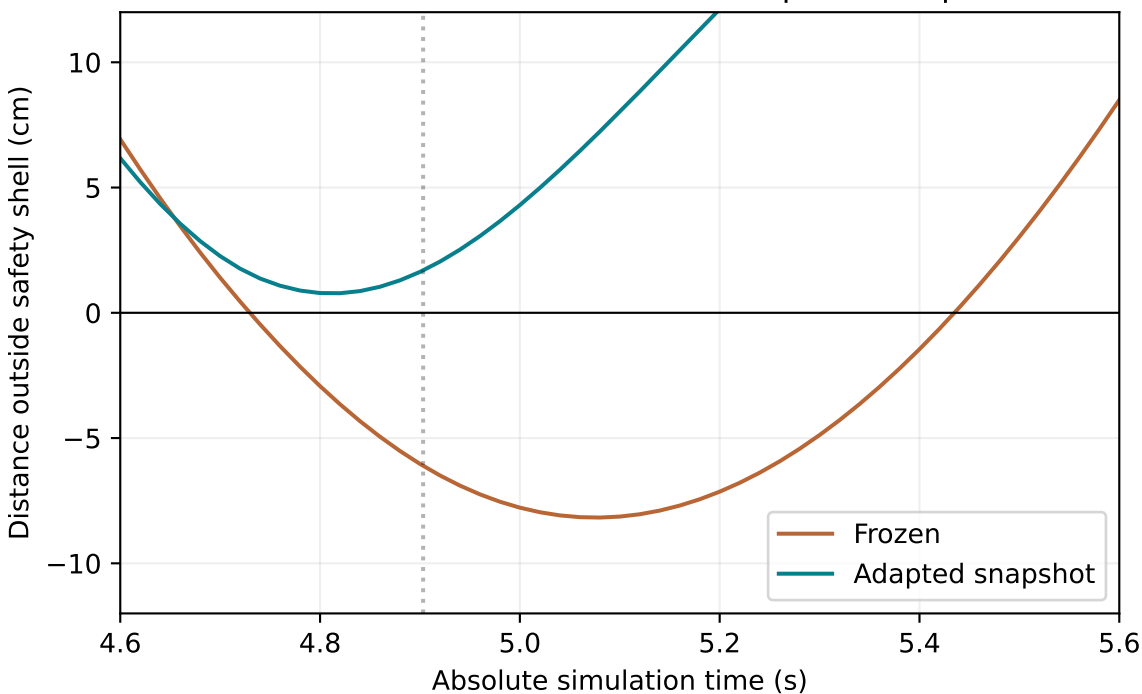
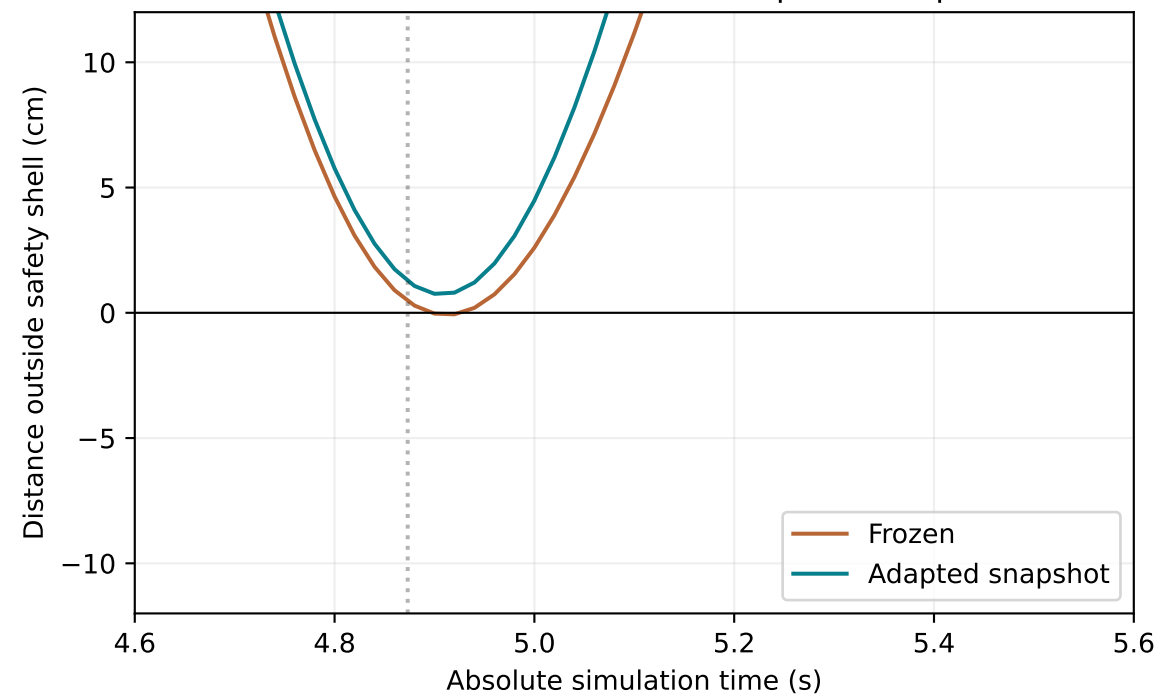


Recovered fallback certificates preserve the requested margin in controlled branches
All physical colliders remain separated; selected cases, not broad superiority

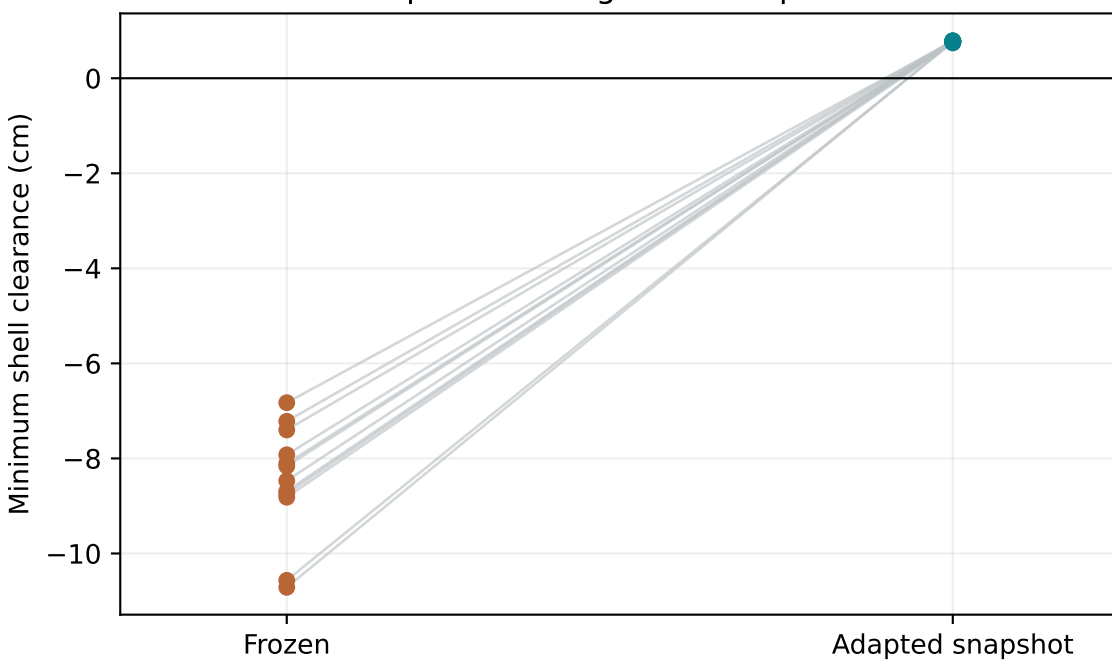
Encounter detail: matched uncompensated pair



Encounter detail: matched compensated pair



12 frozen-protocol neighborhood perturbations



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